

Auteur: Stan Van Campenhout

Studierichting: Informatica

Oefeningenreeks 4A nr. 31.15

$$\begin{aligned}\int \frac{1}{\sqrt{x}-1} dx &\stackrel{x=t^2}{=} \int \frac{1}{\sqrt{t^2}-1} d(t^2) = 2 \int \frac{t}{t-1} dt = 2 \int \frac{t-1+1}{t-1} dt \\ &= 2 \int dt + 2 \int \frac{1}{t-1} d(t-1) = 2t + 2 \ln |t-1| + c \stackrel{t=\sqrt{x}}{=} 2\sqrt{x} + 2 \ln |\sqrt{x}-1| + c\end{aligned}$$

References